

BUILDING A UNIVERSITY TIMETABLE TAKES

~~Weeks.~~

Seconds.

Come see UMTAS live: smarter timetables and campus tools for students, lecturers and admins.

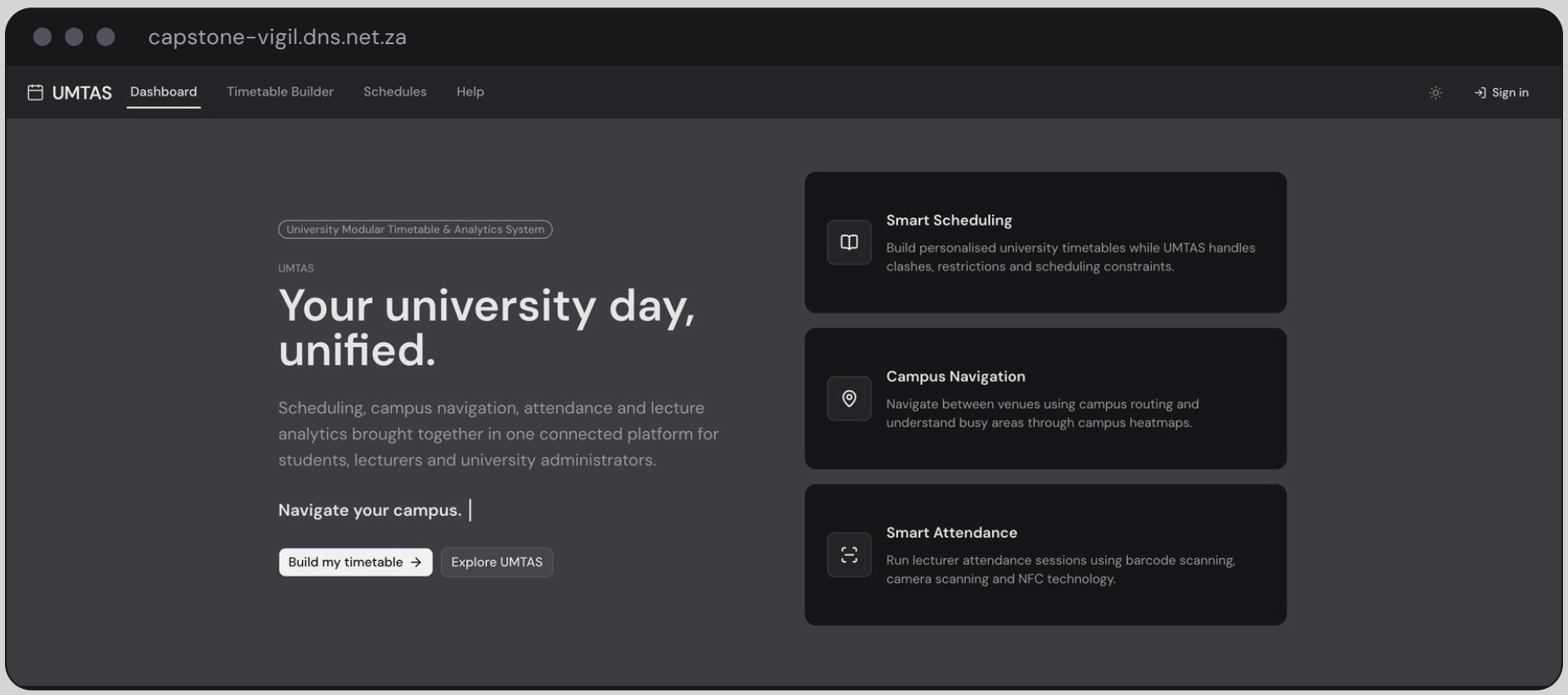
UMTAS

Fri 30 Oct · Rautenbach Hall, UP Hatfield

Swipe →

One platform for campus life.

UMTAS, the University Management & Timetable Analytics System, solves university timetables and helps students, lecturers and administrators manage campus life.



Students

A clash-free week and a route to every class.

Lecturers

Attendance in seconds and a read on the room.

Admins

Timetables, calendars and campus crowds in one place.

Live now at capstone-vigil.dns.net.za

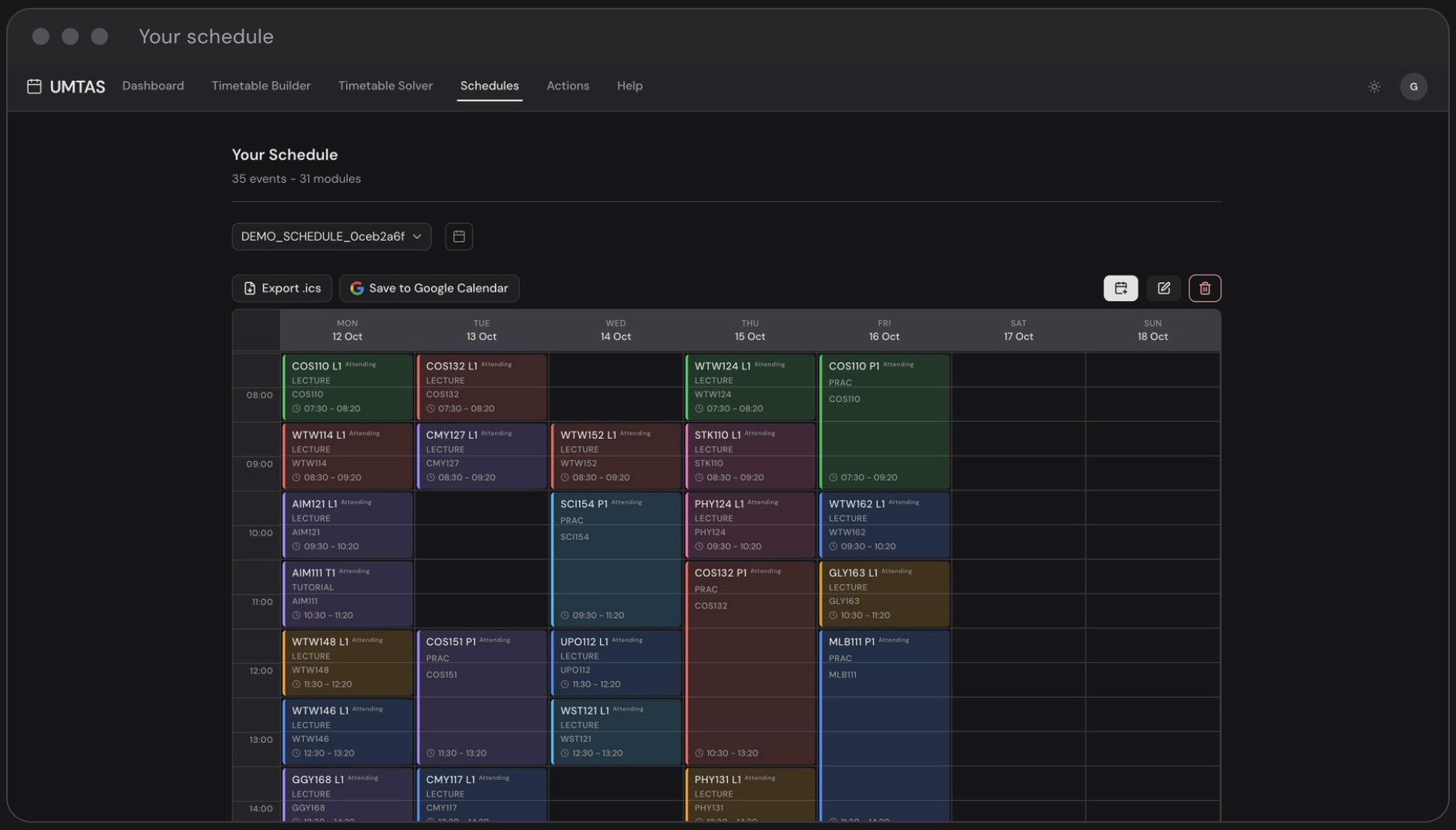
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FEATURE 01

INTELLIGENT TIMETABLE GENERATION

Timetables, solved.

Upload the official timetable PDF, set your preferences, and get a week with no clashes.



Reads your PDF
Modules and events, imported without retyping.

Your preferences
Later starts, a free day, fewer gaps.

Synced
Google Calendar and ICS export.

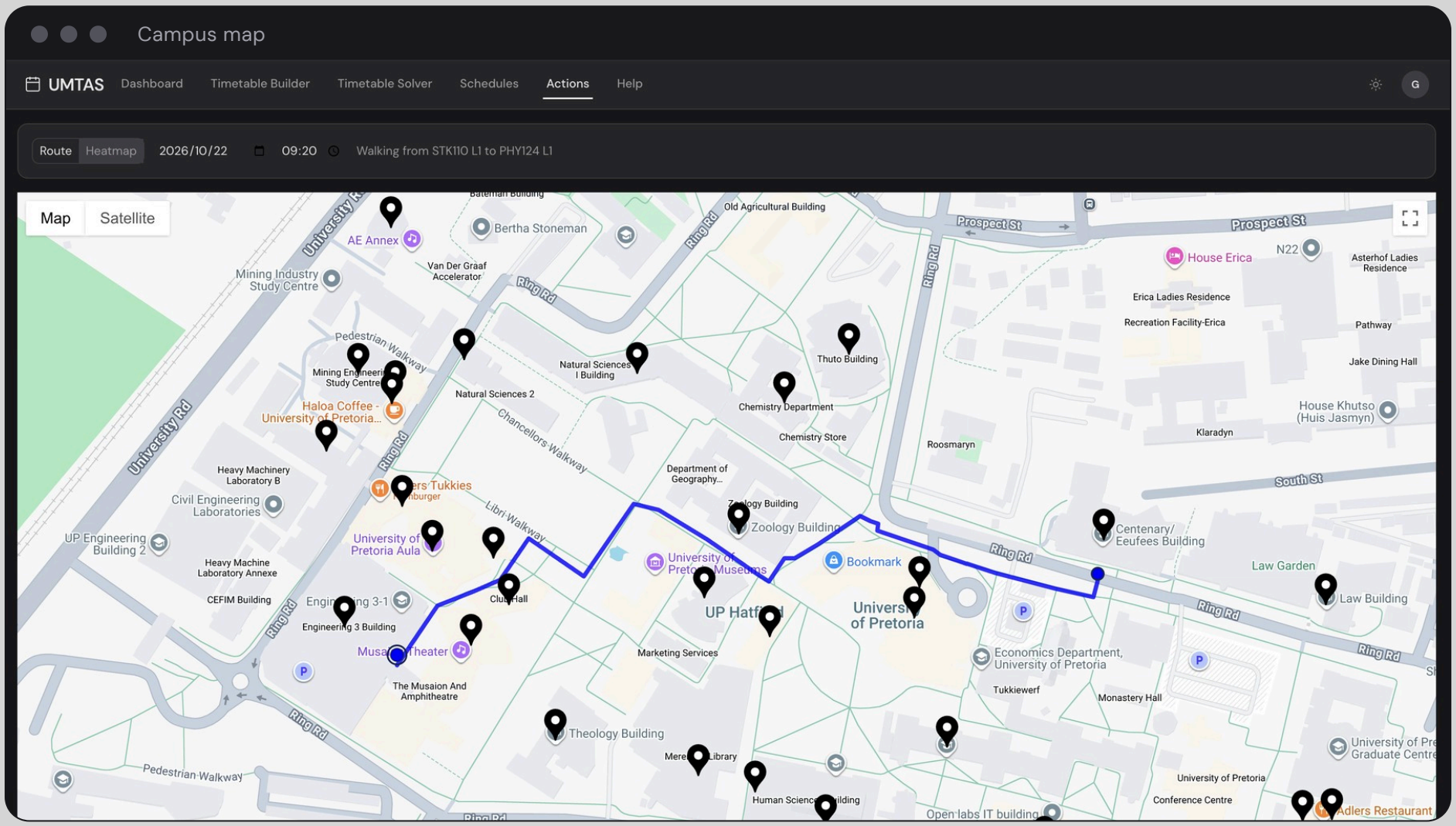
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CAMPUS MAPS, ROUTING & HEATMAPS

Find your way. Skip the crowds.

Walking routes between your classes, and heat maps that show where campus gets busy.



Routes

Directions from one venue to the next.

Heatmaps

Projected foot traffic, hour by hour.

Rerouting

Admins can send students along quieter paths.

UMTAS

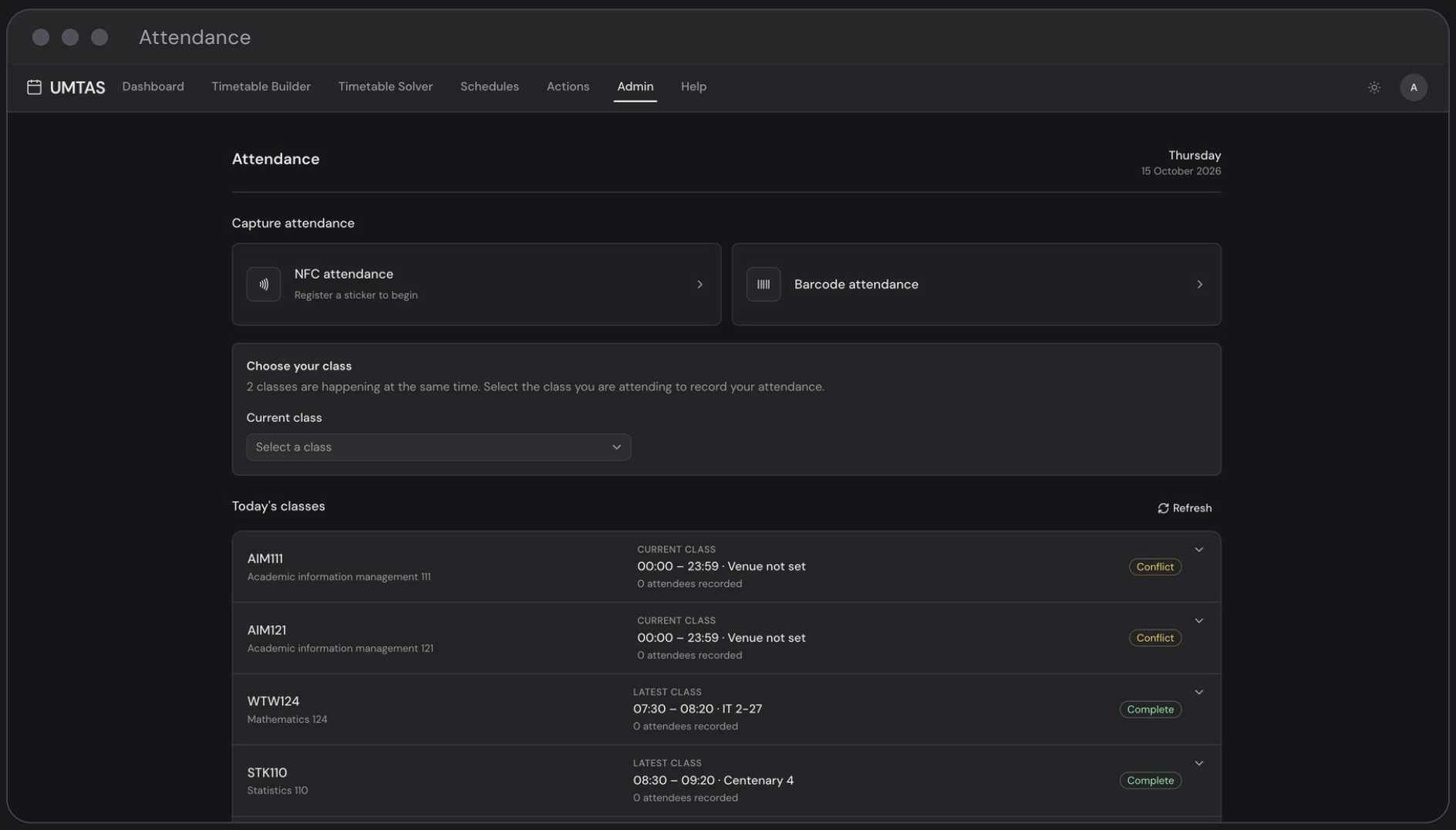
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FEATURE 03

ATTENDANCE TRACKING

Roll call in seconds.

Students check themselves in. Lecturers see who's there, without passing a register around.



Barcode

Scan student cards with a USB scanner.

Camera

Or scan cards with a laptop camera.

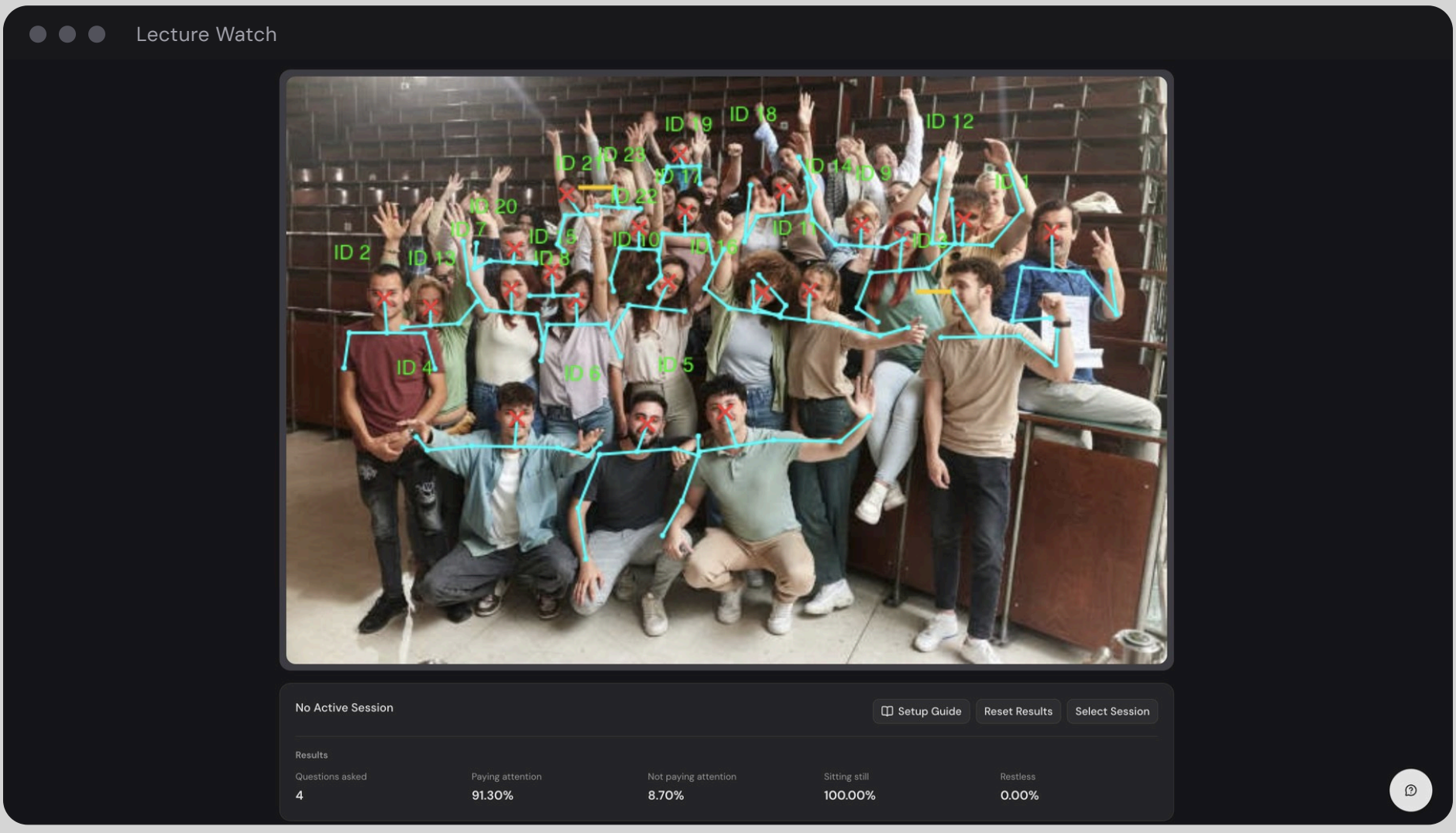
NFC

Students tap the lecturer's sticker with their phone.

LECTURE WATCH

Read the room.

A vision model gives lecturers class-level engagement insights, right in the browser.



Engagement

Raised hands, attention and restlessness.

On-device

Video is analysed in the browser.

Private

Only class-level numbers are saved, never footage.

Built like production software.

A real system that real people log into, tested and monitored like one.

1,100+

automated tests: unit, integration, end-to-end, parser and solver

17/17

quality requirements pass, from speed to security to accessibility

99.97%

public uptime over 30 days

640 ms

95th-percentile response across 40,513 requests at 100 users at once

2 engines

Google OR-Tools CP-SAT, with an evolutionary fallback

3,606

commits in 25 weeks, reviewed through pull requests

Next.js · NestJS · PostgreSQL · Redis · FastAPI · Docker · Grafana

Swipe →

Meet Team Vigil.



Wilmar Smit

TEAM LEAD & INTEGRATION LEAD



Michael Tomlinson

LEAD DEVELOPER & SYSTEM ARCHITECT



Johan Coetzer

FRONTEND LEAD & FULL-STACK DEVELOPER



Marcel Stoltz

DEVOPS LEAD & BACKEND SPECIALIST



Aidan Dawson

BACKEND DEVELOPER & INTEGRATION

WITH THANKS TO OUR INDUSTRY PARTNERS

Tyto Insights & DNS Business

Mentored by Inge Odendaal & Vincent Feistel

SEE IT LIVE

Come see UMTAS live.

Friday, 30 October 2026 · 09:00–13:00

Rautenbach Hall, UP Hatfield Campus

RSVP

Let us know you're coming

Try it now

capstone-vigil.dns.net.za

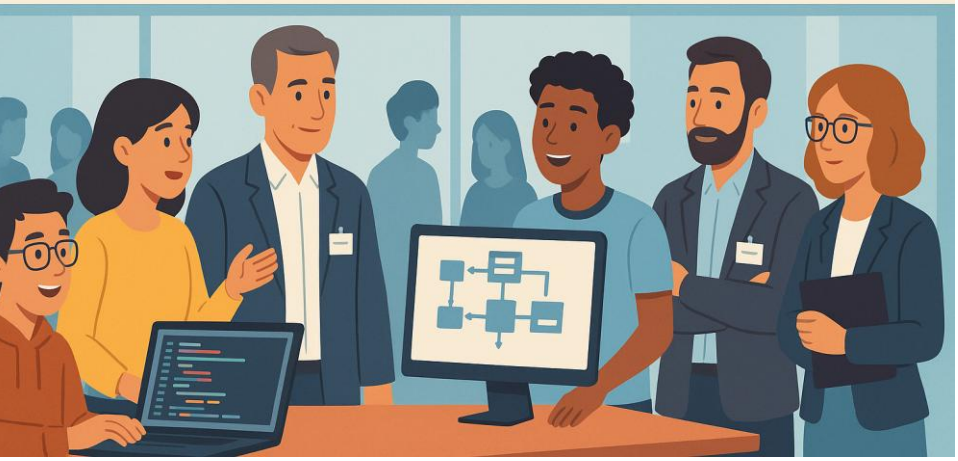
Star the repo

github.com/COS301-SE-2026/UMTAS

UMTAS

Team Vigil · University of Pretoria
COS 301 Capstone 2026

COMPUTER SCIENCE PROJECT DAY



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology
Fakulteit Ingenieurswese, Bou-omgewing en Inligtingtegnologie / Lefapha la Boetšenere, Tikologo ya Kago le Theknolotši ya Tshedimošo

Make today matter
www.up.ac.za

The Head of the Department of Computer Science, University of Pretoria, Professor Hein Venter, cordially invites you to its annual

Computer Science Project Day

On this day, our final-year Computer Science students will present the capstone projects they have developed as the culmination of their academic journey. These projects include web, desktop, and mobile solutions, reflecting the skills and knowledge gained throughout their studies. The event offers a platform to witness the students' creativity, technical expertise, and teamwork, while providing a unique opportunity to engage with them and explore innovative software solutions.

Event Details

Date:	Friday, 30 October 2026
Time:	09:00 to 13:00
Venue:	Rautenbach Hall, Hatfield Campus, University of Pretoria
Parking:	Engineering 3 parkade
GPS:	S25 45' 21"; E28 13' 51"
Dress code:	Semi-formal
RSVP:	By Tuesday 20 October 2026. Please click on the link (https://forms.gle/BrEf8X9gZcV4eQ7b6) or scan the QR Code to RSVP.
Enquiries:	Avinash Singh at asingh@cs.up.ac.za

We would be honoured by your presence at this event, as your engagement will provide valuable encouragement and insight to our students. By interacting with the next generation of computer scientists, you will play a meaningful role in enriching their academic journey and professional growth.

We look forward to welcoming you to this special occasion.

Department of Computer Science Marketing Team

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